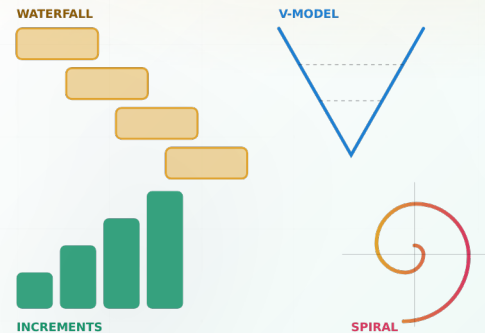


Plan-Driven Process Models I: Waterfall and the V-model and the V-model

Two models that sign things off before moving on. Each is one picture, one rule, and one walk through SANTE-BASSAM — so you can see what the choice of process changes, and what it costs when it is wrong.

**DATE****Tuesday, 15 September 2026****MEETING TIME****1:00 - 2:20 PM****ROOM FORMAT****Lecture + test-writing exercise**

03

2 models worked on SANTE-BASSAM **14** test cases, four levels**1** traceability gap to close **8** figures

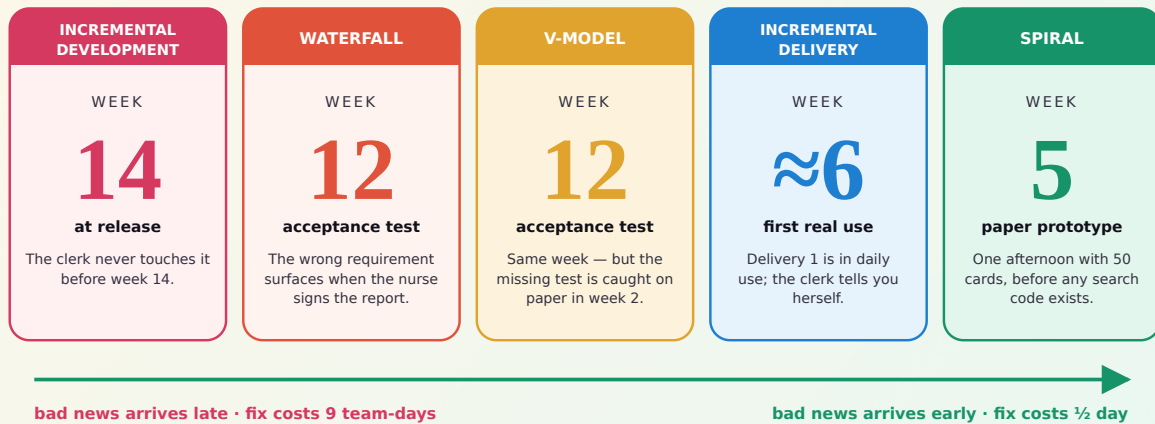
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The session in one picture

Every process model is a different answer to one question: **when does the team find out it was wrong?** This session and the next walk five models through SANTE-BASSAM. Today: the two that sign things off — waterfall and the V-model. Friday: the three that move the risk earlier.

FIG. 0 · SAME TEAM, SAME PROJECT, SAME SURPRISE — ONLY THE WEEK IT ARRIVES CHANGES

"The clerk searches by mother's name." — In which week does the team find out?



A process model is a decision about **when the team is allowed to find out it was wrong**. Everything else in this session is the mechanics of moving that moment to the left.

1 Learning outcomes

L0 3.1 Define waterfall, name its phases in order, and state the four assumptions that make it a rational choice.

L0 3.2 Walk SANTE-BASSAM through each waterfall phase and name the artifact each phase must sign off.

L0 3.3 Explain the V-model as specification levels paired with test levels.

L0 3.4 Write a test at each of the four levels that traces to a named SANTE-BASSAM requirement.

L0 3.5 Read a traceability matrix and find the gap.



How this session is built. Each model: a picture, a rule, then SANTE-BASSAM. Waterfall and the V-model are worked in full today; incremental development, incremental delivery and the spiral are Session 4.

2 The project we translate into



THE SYSTEM

SANTE-BASSAM

Patient registry + appointments for a community health post serving ~6,000 people near Grand-Bassam.



THE PEOPLE

Clerk · nurse · 2 CHWs · district officer · patients

The clerk registers, the nurse runs the post, CHWs do village outreach, the officer needs monthly reports.



THE CONSTRAINTS

Power fails weekly · patchy network

One laptop, one Android tablet. Health-record law fixes the identifier format, consent for minors, retention.



TODAY

Paper

A register, a card per patient, an appointment book, and the nurse's private notebook of follow-ups.



YOUR TEAM

4 students · 14 weeks · no budget

The instructor stands in for the health post when a real stakeholder is unavailable.

FACT 1 · FIXED BY LAW

Identifier format, consent for minors, record retention. These **cannot be discovered by iterating** — you read the regulation once and comply.

FACT 2 · UNKNOWNABLE ON PAPER

The front-desk workflow. Nobody — including the clerk — can describe it precisely **until they see software**.

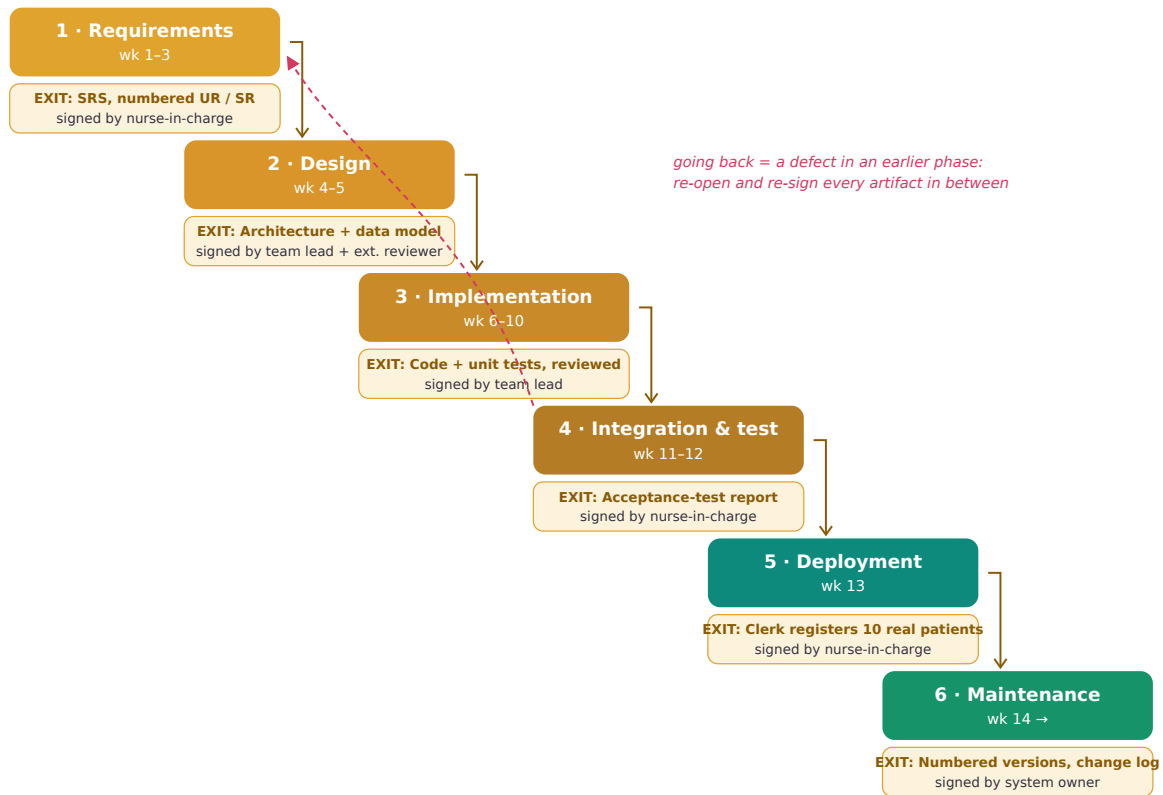
Any process you choose has to accommodate both facts at once. Keep them in view: every model below handles one of them well and the other badly.

3.1 The waterfall model

One rule: each phase finishes with a signed artifact before the next phase begins. Going back is allowed, but treated as a defect, not as the normal way

of working. (Royce, 1970 — named for the shape of the diagram.)

FIG. 1 · WATERFALL, WITH THE SANTE-BASSAM ARTIFACT EACH PHASE MUST SIGN OFF



Read each pale box as an **exit criterion**: the thing that must be true, and signed, before the team may move down a step.

3.1.1.1 What waterfall assumes



Requirements can be known up front

...and stabilised before design.



Late change is expensive

...so front-loading analysis is rational.



The customer can judge a document

...and tell you whether the spec is right.



Hand-offs beat conversation

...because the team is large or spread across organisations.

ASSUMPTIONS HOLD →

Regulated / safety-critical work; contract development where scope defines payment; systems integration with interfaces fixed a year ahead. **Waterfall is a sound choice.**

ASSUMPTIONS FAIL →

Whenever the customer will only recognise what they want *by seeing it*. Waterfall then discovers the wrong requirement at the most expensive moment: **acceptance testing**.

APPLIED TO SANTE-BASSAM

Fourteen weeks of pure waterfall — and the week-12 discovery

FIG. 2 · THE PLAN (TOP) AND THE SURPRISE (BOTTOM)

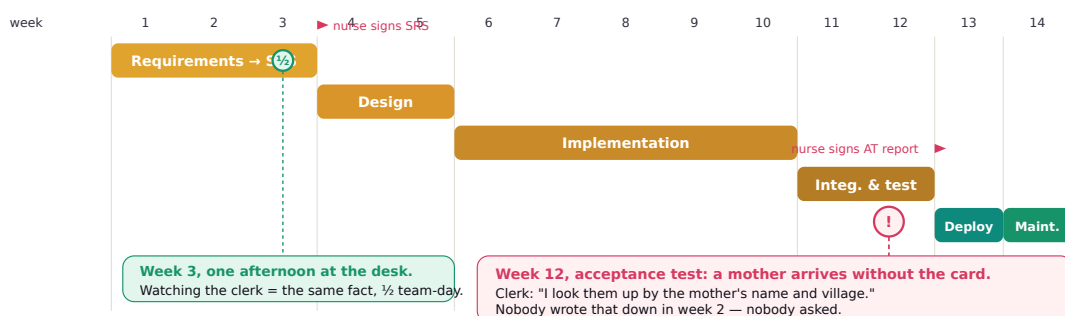


FIG. 3 · WHAT THE SAME REQUIREMENT COSTS IN WEEK 12 VS WEEK 3 (TEAM-DAYS)



WHAT WATERFALL BUYS

An **auditable trail**. If the Ministry asks which test proves a power-cut record is not lost: the SRS names UR-2, the report names AT-02, and the nurse's signature is on both. A health-record system is obliged to produce exactly this evidence.

WHAT WATERFALL BETS

That week-3 elicitation is good enough that week-12 surprises are rare. **Safe for the parts fixed by law. Not safe for the front-desk workflow.**

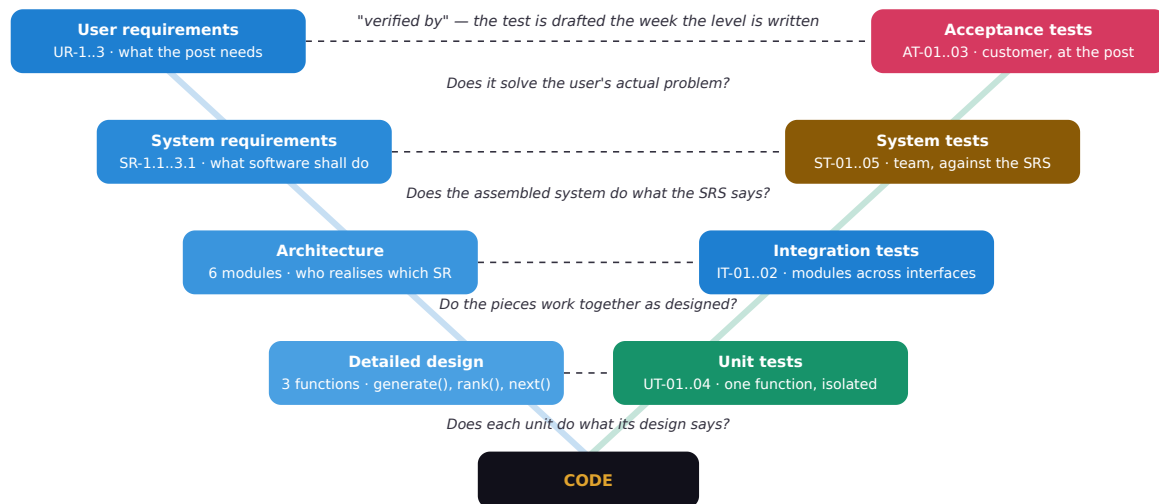


Carry forward. Waterfall converts uncertainty about requirements into one large risk that lands at the end. Every other model in this session is a different way of moving that risk earlier.

3.2 The V-model: specification and test as mirror images

One rule: every level of specification on the left arm is verified by exactly one level of test on the right — and the test is *written when the level is specified*, not when the code exists.

FIG. 4 · THE V, LABELLED WITH THE SANTE-BASSAM TESTS WE WRITE AT EVERY LEVEL



The payoff is **bidirectional traceability**: for any test, which requirement? For any requirement, which test? A requirement with no test is unverified; a test with no requirement is waste or a hidden requirement. Both are defects, and the V makes them visible — which is why it dominates medical-device and avionics work.

APPLIED TO SANTE-BASSAM

Three user requirements, refined all the way down, then tested all the way up

LEFT ARM · LEVEL 1 — USER REQUIREMENTS

UR-1 The clerk can find an existing patient's record in under one minute, even with no card.

UR-2 A record captured during a power cut or offline is never lost.

UR-3 The district officer gets the monthly report without anyone re-counting the paper register.

LEFT ARM · LEVEL 2 — SYSTEM REQUIREMENTS

SR-1.1 Find patients by card number, by family + given name, or by mother's name + village.

SR-1.2 Name search tolerates spelling variants; candidates ranked, with village and birth year shown.

SR-1.3 First page of a 10,000-record search within 2 s on the target laptop.

SR-2.1 Every registration and visit persisted locally *before* the confirmation is shown.

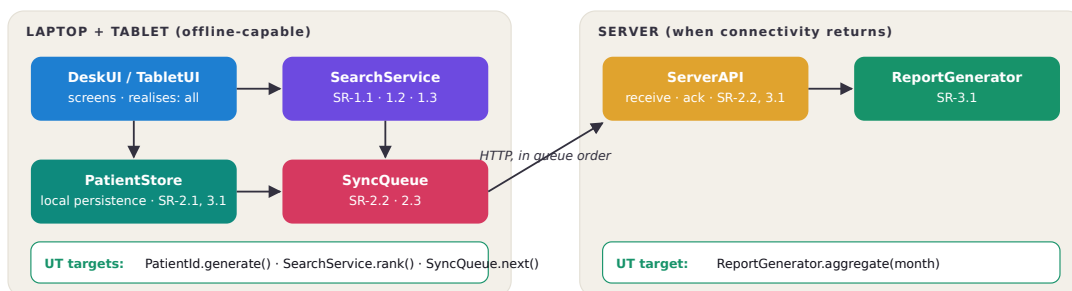
SR-2.2 Offline records marked *pending*, uploaded in creation order, *synced* only after server acknowledgement.

SR-2.3 A record failing upload three times is flagged for supervisor review and kept on the device.

SR-3.1 Monthly report of visits by outcome (attended / missed / rescheduled) per village.

LEFT ARM · LEVELS 3 & 4 — ARCHITECTURE AND DETAILED DESIGN

FIG. 5 · SIX MODULES, WHICH SR EACH REALISES, AND THE THREE FUNCTIONS WE UNIT-TEST



`generate()` → ids like `HP-BSM-2026-XXXXXX`, unique across both devices with no network · `rank()` → 0–100 by name similarity, +15 village match, +10 birth year within one · `next()` → oldest pending; after 3 failures move to *needs-review* and continue.

RIGHT ARM — ONE FULLY WRITTEN TEST PER LEVEL (THE OTHER TEN FOLLOW THE SAME SHAPE)

AT-01 Find a returning patient who has no card

verifies UR-1

- GIVEN** the register is migrated; a mother arrives with her child but no card
- WHEN** the clerk asks only what she would normally ask (child's name, mother's name, village)
- THEN** the correct record opens within 60 s in ≥ 9 of 10 trials with real returning patients

ST-03 Offline capture, ordered sync, acknowledgement

verifies SR-2.1, 2.2

- GIVEN** the tablet in airplane mode
- WHEN** three patients are registered, then connectivity returns for 30 s
- THEN** each shows *pending* on save; server receives them in creation order; each becomes *synced* only after the ack is logged

IT-02 SyncQueue ↔ ServerAPI

verifies the sync contract

- GIVEN** 5 pending records; ServerAPI running locally
- WHEN** sync is triggered
- THEN** 5 POSTs arrive in queue order; each ack carries the record id; SyncQueue marks synced in the same order

UT-03 SyncQueue.next() ordering and give-up rule

verifies detailed design

- GIVEN** A, B, C enqueued in order; A fails every attempt
- WHEN** next() is called repeatedly, failures recorded
- THEN** A returned 3x, then B; A's status is *needs-review*; C follows B

AT-02 power cut mid-registration → complete or absent, never half-saved

AT-03 report equals the officer's own hand count; she'd submit it

ST-01 "Aya Kofi" + Modeste → 7 Modeste matches first

ST-02 p95 of 50 searches ≤ 2.0 s

ST-04 3 failed uploads → needs-review, still readable

ST-05 12 report cells match the seed; total 164

IT-01 DeskUI → SearchService → PatientStore, one query, same list

UT-01 200,000 ids on two devices, all distinct, all match the pattern

UT-02 rank() scores strictly descending, first ≥ 90

UT-04 aggregate() equals the hand-computed 3×3 matrix

THE TRACEABILITY MATRIX — THE ARTIFACT THE V EXISTS TO PRODUCE

FIG. 6 · REQUIREMENT × TEST-LEVEL COVERAGE

	AT	ST	IT	UT	status
UR-1 → SR-1.1, 1.2, 1.3	AT-01	ST-01, 02	IT-01	UT-02	covered
UR-2 → SR-2.1, 2.2, 2.3	AT-02	ST-03, 04	IT-02	UT-01, 03	covered
UR-3 → SR-3.1	AT-03	ST-05	— GAP —	UT-04	no IT

Reading the last row is the whole skill. UR-3 has no integration test: nobody has checked that ReportGenerator reads the same Visit records SyncQueue wrote. Found in week 2 by looking at a table — five minutes. Found in week 13 — a wrong report on the district officer's desk.



Common student error: writing all four levels as variations of one scenario. Look down the UR-2 column: AT-02 pulls a plug in front of the nurse; ST-03 scripts airplane mode; IT-02 checks the order of HTTP calls; UT-03 checks a queue rule with no I/O at all. **Each level answers a different question and catches a different defect.**

4

In-class exercise · close the gap

Groups of four, 15 minutes. The traceability matrix in Fig. 6 shows UR-3 has no integration test. Fix the test plan, then extend it.



Write IT-03

— ReportGenerator reading the Visit records SyncQueue wrote:
Given / When / Then, in the format of the four worked tests. Say which two modules it exercises together.



Add a fourth user requirement

— UR-4: appointments (from S-10).
Refine it into two system requirements and name the module that realises each.



Fill the row

— one test at each of the four levels for UR-4, each answering a *different* question (re-read the warning callout before you start).



Defend one test

— which of your four would catch a defect none of the other three could? Two minutes to present.



Expect this. Most groups write the AT and the ST as the same scenario at two speeds. The IT is the one that forces you to name an interface; the UT is the one with no I/O at all.

5

Homework • due before Session 4 (Friday 18 September)



Reading

— Sommerville 10th ed., §2.1 (waterfall, incremental) and §2.2 (process activities). Come with the definition of *incremental delivery* written in your own words — Session 4 leaves it for you to work.



Traceability

`s03-traceability.md` — the matrix of Fig. 6 with your IT-03 and the UR-4 row from today's exercise, pushed to the team repo.



Assessment link. The four test levels become the test-plan section of your Project Plan (due Session 10).

TWO MODELS · ONE RULE EACH · WHAT EACH MOVES EARLIER

WATERFALL

sign off each phase before the next

Moves nothing earlier — the risk lands at acceptance test, week 12, at 18× the cost. Buys an auditable trail the Ministry can read.

V-MODEL

write the test when you write the spec

Moves *test-plan gaps* earlier: UR-3's missing IT found on paper in week 2. Use where evidence must be signed.

FRIDAY

incremental dev · incremental delivery · spiral

Three ways of moving the requirements risk itself — and a decision heuristic for choosing.

If you remember one sentence: **waterfall converts uncertainty about requirements into one large risk that lands at the end.**

Key terms

waterfall model

phase sign-off

exit criterion

V-model

acceptance / system / integration / unit test

bidirectional traceability

traceability matrix

Readings and references

1. Sommerville, I. *Software Engineering*, 10th ed. — Chapter 2, §2.1–2.3.
2. Royce, W. *Managing the Development of Large Software Systems*. Proc. IEEE WESCON, 1970.

3. Forsberg, K. & Mooz, H. *The Relationship of System Engineering to the Project Cycle* (the V-model). NCOSE, 1991.
4. Pressman & Maxim, *Software Engineering: A Practitioner's Approach* — Chapter 2.

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Next · Session 04 — Plan-Driven Process Models II: Increments, the Spiral and Choosing